📌 DATE and TIME functions in MYSQL

1. Functions to Get Current Date and Time

These functions retrieve the current date, time, or both, as perceived by the database server.

Function	Description	Example Query	Example Output (Approximate)
`NOW()`	Returns the current date and time.	`SELECT NOW();`	`2025-01-01 09:28:00`
`CURDATE()`	Returns the current date only.	`SELECT CURDATE();`	`2025-01-01`
`CURTIME()`	Returns the current time only.	`SELECT CURTIME();`	`09:28:00`

2. Functions for Date Arithmetic

These functions allow you to add or subtract intervals from a given date.

Function	Description	Example Query	Example Output
`DATE_ADD()`	Adds a specified time interval to a date.	`SELECT DATE_ADD('2025-01-01', INTERVAL 30 DAY);`	`2025-01-31`
`DATE_SUB()`	Subtracts a specified time interval from a date.	`SELECT DATE_SUB('2025-01-01', INTERVAL 2 MONTH);`	`2024-11-01`
`DATEDIFF()`	Returns the number of days between two dates.	`SELECT DATEDIFF('2025-12-25', '2025-01-01');`	`358`
`TIMESTAMPDIFF()`	Returns the difference between two timestamps in	`SELECT TIMESTAMPDIFF(HOUR,`	`5`

	a specified unit (e.g., `HOUR`, `MONTH`).	`'2025-01-01 10:00:00', '2025-01-01 15:00:00');`

3. Functions to Extract Date Parts

These functions extract specific components (e.g., year, month, day) from a date or datetime value.

Function	Description	Example Query	Example Output
`YEAR()`	Extracts the year.	`SELECT YEAR('2025-01-01');`	`2025`
`MONTH()`	Extracts the month (numeric).	`SELECT MONTH('2025-01-01');`	`1`
`DAY()`	Extracts the day of the month.	`SELECT DAY('2025-01-01');`	`1`
`DAYNAME()`	Returns the name of the weekday.	`SELECT DAYNAME('2025-01-01');`	`Wednesday`
`EXTRACT()`	A general-purpose function to extract a specific unit.	`SELECT EXTRACT(QUARTER FROM '2025-01-01');`	`1`
`LAST_DAY()`	Returns the date of the last day of the month.	`SELECT LAST_DAY('2025-02-10');`	`2025-02-28`

4. Functions for Formatting and Conversion

These functions change the display format of a date or convert strings into date types.

Function	Description	Example Query	Example Output
`DATE_FORMAT()`	Formats a date into a custom string format.	`SELECT DATE_FORMAT('2025-01-01', '%W, %M %d, %Y');`	`Wednesday, January 01, 2025`

`STR_TO_DATE()`	Converts a string into a date based on a format pattern.	`SELECT STR_TO_DATE('01-01-2025', '%d-%m-%Y');`	`2025-01-01`

Summary Table of Format Specifiers

When using `DATE_FORMAT()`, use these common specifiers:

Specifier	Description	Example
`%Y`	4-digit Year	2026
`%m`	Month (01-12)	01
`%M`	Month Name	January
`%d`	Day of Month (01-31)	01
`%W`	Weekday Name	Thursday

`DAYOFWEEK(date)`

This function returns the weekday index for a specified date, ranging from 1 to 7. The index starts with Sunday as 1.

- **Syntax:** `DAYOFWEEK(date)`
- **Returns:** An integer representing the day of the week (1=Sunday, 2=Monday, ..., 7=Saturday).

Example Query	Output	Explanation
`SELECT DAYOFWEEK('2025-01-20');`	`2`	January 20, 2025 is a Monday, which is the second day of the week (Sunday is 1).
`SELECT DAYOFWEEK(CURDATE());`	(varies)	Returns the weekday index for the current server system date.

`DAYOFYEAR(date)`

This function returns the day of the year for a given date, ranging from 1 to 366 (for a leap year).

- **Syntax:** `DAYOFYEAR(date)`
- **Returns:** An integer representing the ordinal day of the year (1 for January 1st).

Example Query	Output	Explanation
`SELECT DAYOFYEAR('2025-01-20');`	`20`	January 20th is the 20th day of the year.
`SELECT DAYOFYEAR('2025-12-31');`	`365`	December 31st is the last day of the year in 2025 (not a leap year).

`WEEKOFYEAR(date)`

This function returns the calendar week number of the year for a given date, with the range typically being 1 to 53. By default, it follows the ISO 8601 standard, where the first week of the year is the one that contains the first Thursday.

- **Syntax:** `WEEKOFYEAR(date)`
- **Returns:** An integer representing the week number of the year.

Example Query	Output	Explanation
`SELECT WEEKOFYEAR('2025-01-20');`	`4`	January 20, 2025 falls into the fourth full week of the year according to ISO 8601 rules.
`SELECT WEEKOFYEAR('2025-01-01');`	`1`	January 1st, 2025 (a Wednesday) is part of the first week of the year under the ISO 8601 standard.

`HOUR()` Function

The `HOUR()` function returns the hour as an integer. For time-of-day values, the range is 0 to 23. However, if the expression represents a time interval that exceeds 24 hours (up to '838:59:59'), the returned value can be greater than 23.

SQL Example	Output	Explanation

`SELECT HOUR('10:30:45');`	`10`	Extracts the hour from the time string.
`SELECT HOUR('2025-06-01 10:00:00');`	`10`	Extracts the hour from the `DATETIME` value.
`SELECT HOUR(CURTIME());`	*Current Hour (e.g., 9)*	Extracts the hour from the current system time.

`MINUTE()` Function

The `MINUTE()` function returns the minute as an integer, ranging from 0 to 59.

SQL Example	Output	Explanation
`SELECT MINUTE('10:30:45');`	`30`	Extracts the minute from the time string.
`SELECT MINUTE('2025-06-01 10:00:00');`	`0`	Extracts the minute from the `DATETIME` value.
`SELECT MINUTE('4:10:17');`	`10`	Extracts the minute from a time value where a single digit is used for hours.

`SECOND()` Function

The `SECOND()` function returns the second as an integer, ranging from 0 to 59.

SQL Example	Output	Explanation
`SELECT SECOND('10:30:45');`	`45`	Extracts the second from the time string.
`SELECT SECOND('2025-06-01 10:00:00');`	`0`	Extracts the second from the `DATETIME` value.

EXTRACT() in mysql

Syntax:

```
EXTRACT(unit FROM date)
```

- `unit`: Specifies the part of the date/datetime to extract (e.g., `YEAR`, `MONTH`, `DAY`, `HOUR`, `MINUTE`, `SECOND`, `MICROSECOND`, etc.).
- `date`: The `DATE`, `DATETIME`, or `TIMESTAMP` value from which to extract the part.

Examples and Outputs

Here are several examples of using the `EXTRACT()` function with their corresponding outputs:

Query	Expected Output	Explanation
`SELECT EXTRACT(YEAR FROM '2024-09-22');`	`2024`	Extracts the year from the date value.
`SELECT EXTRACT(MONTH FROM '2024-09-22');`	`9`	Extracts the month (as a number) from the date value.
`SELECT EXTRACT(DAY FROM '2024-09-22');`	`22`	Extracts the day of the month.
`SELECT EXTRACT(QUARTER FROM '2024-09-22');`	`3`	Extracts the quarter of the year (1-4).
`SELECT EXTRACT(WEEK FROM '2024-09-22');`	`38`	Extracts the week number of the year.
`SELECT EXTRACT(HOUR FROM '2024-09-22 15:30:45');`	`15`	Extracts the hour from the datetime value.
`SELECT EXTRACT(MINUTE FROM '2024-09-22 15:30:45');`	`30`	Extracts the minute from the datetime value.
`SELECT EXTRACT(SECOND FROM '2024-09-22 15:30:45');`	`45`	Extracts the second from the datetime value.
`SELECT EXTRACT(YEAR_MONTH FROM '2024-09-22 15:30:45');`	`202409`	Extracts the year and month as a combined integer (YYYYMM).
`SELECT EXTRACT(DAY_HOUR FROM '2024-09-22 15:30:45');`	`2215`	Extracts the day and hour as a combined integer (DDHH).

`SELECT EXTRACT(MINUTE_SECOND FROM '2024-09-22 15:30:45');`	`3045`	Extracts the minute and second as a combined integer (MMSS).